

Figuring Out A Research Question:

-How can we use the knowledge of the vascular system to further elaborate/interpret neuroimaging studies? (General question/Aim) (The impact of the vascular system on the future of neuroimaging studies)

- How can further studies into vascular imaging allow us to further interpret neuroimaging studies?
- How do changes in the vascular system impact 'neural networks'?
- How can we indicate changes in the 'neural networks' with the use of the vascular system through ASL & BOLD fMRI? (more specific)

- Can we explain the difference between vascular-driven connectivity and neurally driven connectivity? (Yes)

- At what spatial resolution (0.35mm vs. 7mm) are we able to see an operational distinction between neuronal and vascular-driven signals?

*Can differences in latency or frequency help separate vascular and neuronal connectivity?

- Can this operational distinction be seen clearly/occur during the Simon task?
- Can this operational distinction be seen clearly/occur during the resting task?
- How do the temporal profiles of vascular-driven signals differ from neuronal-driven signals?-somewhat related
- **Can vascular connectivity arise independently of neuronal activity?**

-Can we separate brain connectivity into a "vascular" network and a "neuronal" network using fMRI data? I expect..

- Do vascular and neuronal connectivity differ in timing?
- Can we still see vascular connectivity effects in normal fMRI scans without a CO₂ task? (use of breathhold/resting task)

Main research question:

-Can we separate brain connectivity into a "vascular" network and a "neuronal" network using fMRI data?

Sub-question in terms of "space":

-Does the spatial structure of vascular-driven connectivity match the vascular anatomy we know?

Sub-question in terms of "time":

- Do vascular and neuronal connectivity differ in timing?
- How do the temporal profiles of vascular-driven signals differ from neuronal-driven signals?
- Can vascular-driven connectivity arise independently of neuronal activity (or vice versa)?

Sub-question in terms of "tasks":

- Can this operational distinction be seen clearly/occur during the Simon task?

- Can this operational distinction be seen clearly/occur during the resting task?

*It's more of a hypothesis, how you got there, what are your plans to test it in plain English (e.g. "I run a comparison"), and if you want and can, an expectation on the possible outcomes. Then the methods is where you state technically how you will do things (i.e. comparison → T-test)

Answering The Research Questions:

- An ICA will be conducted to separate brain signals into separate 'networks': a neuronal 'network' (which will be shown through tasks such as the Simon task) and a Vascular 'network' (which will be shown through tasks such as the breath-hold task).
- A Normalized R2 will be conducted to prove these two 'networks' are different, and will be calculated to signal how much effort comes from each respective 'network'.
- A standard t-test will be used to confirm the statistical distinction of these 'networks.'
- Finally, a 3-way ANOVA will be used to see if these network patterns hold up across all participants

Presentation Notes:

- Information listed below is relevant to the respective slide it falls under

Slide 4

The component with the maximal negative correlation with the 3-back stimulus was identified as the neuronal Default Mode Network (DMN)

In these results, we identified that **the Default Mode Network (DMN)** was represented by two components: one DMN time-series showed clear BOLD signal increases lagging the end-tidal CO2 effects, thereby reflecting the vasodilatory effect of the stimulus. The second DMN component time-series exhibited BOLD signal decreases during and preceding the actual breath-hold itself, potentially reflecting deactivation of this network and reduced neural activity during the active, attentional portion of the paradigm.

Slide 5

The component demonstrating maximum positive correlation with the 3-back stimulus was identified as the neuronal Task Positive Network (TPN) (Fox et al., 2005; Spreng, 2012).

Slide 6

Finally, the component exhibiting maximum positive correlation with the visual stimulus was identified as the neuronal Visual Network (VN).

Slide 8

Moore and Cao: hemodynamics also play a role in information processing through modulation of neural activity. Functional hyperemia, the basis of the functional MRI (fMRI) BOLD signal, is a localized influx of blood correlated with neural activity levels. Functional

hyperemia is considered by many to be excessive from a metabolic standpoint, but may be appropriate if interpreted as having an activity-dependent neuro-modulatory function.

Slide 10

Resting State Within the resting state task participants were instructed to keep their eyes open and fixate on a cross in the middle of a black screen while staying as still as possible. The BOLD signal records spontaneous fluctuations in hemodynamics (Chen et al., 2020).

Simon Task Within the Simon participants are instructed to look at a screen where red or green squares appear on either the left or right side. They are instructed to press a button with their right hand for red and their left hand for green as fast as they could. With Congruent trials in which the square appears on the same side as the hand used. And incongruent trials in which the square appears on the opposite side. Within this task 48 stimuli were congruent and 48 were incongruent. Every response was recorded, and indications were to be made as to whether the response was correct or incorrect. This task is used to identify components that exhibit the strongest positive correlation with a goal-directed neuronal stimulus (Bright et al., 2020). **Breath-Hold task** The breath hold task serves as a controlled stimulus to drive cerebrovascular reactivity and facilitate the "splitting" of networks into neuronal and vascular pairs. Within this task participants alternate between periods of normal breathing and short-duration breath holds. Which were composed of four paced breathing cycles of 6 seconds each, a breath hold period of 20 seconds, followed by an exhalation of 3 seconds, and finally a period of 11 seconds of recovery breathing. The total duration of this task was 6 minutes. This manifests as a global BOLD signal increase across the brain, which can be modeled using the subject-specific trace to identify vascular 'network' components (Bright et al., 2020).

Slide 11

An (ICA) Independent Component Analysis will be used to separate brain signals into separate 'networks': a neuronal 'network' (which will be shown through tasks such as the Simon task) and a Vascular 'network' (which will be shown through tasks such as the breath hold task). Distinguishing the Neuronal 'network' by the signals that match the task timing and the Vascular 'network' by seeing which match CO₂ levels. To prove these two 'networks' are different, **A Normalized R²** will be calculated to signal how much effort comes from each respective 'network'. Then a standard **comparison test (t-test)** will be used to confirm the statistical distinction of these 'networks.' Finally, a group **test (3-way ANOVA)** will be used to see if these network patterns hold up across all participants.

Simplifying Code Into English Text:

- Step 1: Start with the data (integrate the data into Bash)
- Step 2: Preprocessing (Filter/double-check the data)
- Step 3: Run MELODIC
- Step 4: Load data from MELODIC into Python

- Step 5: Load the ICA time series
- Step 6: Now use the ICA data to compare against CO₂ data (repeat for every ICA component)
- Step 7: Calculate the lag (possible delay due to CO₂)
- Step 8: Spatial Overlap? (used to test whether or not the data has a strong overlap)
- Step 9: Classify Components
 - Break down the data and explain what can be inferred from it
- Step 10: Statistical testing (Neuronal vs. Vascular - t-test/ANOVA)